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Fabric resistant to liquid penetration

Abstract

Described embodiments generally relate to a garment for resisting penetration of a liquid. The garment comprises an outer layer, and an inner layer. The inner layer is configured to be positioned closer to the skin of a wearer than the outer layer when the garment is being worn. Each of the inner layer and the outer layer on their own are wettable by the liquid, and the inner layer has a lower wettability to the liquid than the outer layer. The configuration of inner and outer layers creates a wettability difference that biases the outer layer to absorb the liquid and the inner layer to resist penetration of the liquid in a direction moving from the outer layer to the inner layer.

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Background/Summary

RELATED APPLICATIONS
(1) This patent arises from the U.S. national stage of International Patent Application Serial No. PCT/AU2021/050402, having an international filing date of May 3, 2021, which is hereby incorporated by reference in its entirety. Further, this patent claims priority to Australian Patent Application No. 2020901722, which was filed on May 27, 2020.
TECHNICAL FIELD
(2) Embodiments generally relate to fabrics and garments that are resistant to liquid penetration, as well as to methods for the production of such fabrics and garments.
BACKGROUND
(3) Liquid resistant fabrics are often required in scenarios relating to keeping people or products dry or protected from particular liquids. For example, in the field of protective clothing, fabrics that are resistant to liquid chemicals are highly desirable. Exposure to toxic industrial chemicals (TICs) and chemical warfare agents (CWAs) can cause health problems or even lead to death of humans and animals. To protect people from contacting the hazardous liquids, protective clothing is essential. Semi-permeable and impermeable materials provide good barriers to penetration of chemical agents in liquid, vapor, and aerosol form, but can lead to serious heat stress consequences for users. Air and moisture permeable fabrics are preferred for their breathability and comfort.
(4) Currently, chemical protective clothing (CPC) is designed and manufactured based on the principle of maximum possible repellency and absorption. Specifically, CPC is often designed with

the outer shell fabric coated with a liquid repellent coating to offer liquid repellency, while an inner fabric that is designed to contact the skin is often an absorptive material providing liquid and gas absorption. However, current CPC is not capable of repelling hazardous chemicals with a low surface tension. Extensive efforts have been made to increase the repellency of the outer shell layer and the absorption or thickness of the inner layer, but these methods have had limited success while maintaining air-permeability, which is important for maintaining comfort for the wearer. No one fabric is able to repel all liquids.

(5) It is desired to address or ameliorate one or more shortcomings or disadvantages associated with prior fabrics that are resistant to liquid penetration, or to at least provide a useful alternative thereto.

(6) Throughout this specification the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or group of elements, integers or steps.

(7) In this document, a statement that an element may be “at least one of” a list of options is to be understood to mean that the element may be any one of the listed options, or may be any combination of two or more of the listed options.

(8) Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each of the appended claims.

SUMMARY

(9) Some embodiments relate to a garment for resisting penetration of a liquid, the garment comprising: an outer layer; and an inner layer; wherein the inner layer is configured to be positioned closer to the skin of a wearer than the outer layer when the garment is being worn; wherein each of the inner layer and the outer layer on their own are wettable by the liquid; wherein the inner layer has a lower wettability to the liquid than the outer layer; and wherein the configuration of inner and outer layers creates a wettability difference that biases the outer layer to absorb the liquid and the inner layer to resist penetration of the liquid in a direction moving from the outer layer to the inner layer.

(10) In some embodiments, when worn with the inner layer positioned closer to the skin of the wearer than the outer layer, the garment provides greater resistance to penetration of the liquid toward the skin of the wearer than either a garment made from either the inner layer or the outer layer on its own, or than a garment worn with the outer layer positioned closer to the skin of a wearer than the inner layer.

(11) According to some embodiments, the liquid is a low surface tension liquid. In some embodiments, the surface tension of the liquid is less than 50 mMm.sup.−1. In some embodiments, the surface tension of the liquid is less than 40 mMm.sup.−1. According to some embodiments, the surface tension of the liquid is less than 30 mMm.sup.−1.

(12) In some embodiments, the garment is air-permeable.

(13) According to some embodiments, the inner layer and the outer layer both repel high surface tension liquids. In some embodiments, the inner layer and the outer layer both absorb high surface tension liquids. In some embodiments, the inner layer repels high surface tension liquids. According to some embodiments, the outer layer absorbs high surface tension liquids. In some embodiments, the absorption layer and the resistant layer both comprise at least one of a hydrophobic or super hydrophobic material.

(14) In some embodiments, the inner layer comprises a coating. In some embodiments, the outer layer comprises a coating. According to some embodiments, the coating is applied using a dip coating method. According to some embodiments, the coating is applied using a spray coating method. In some embodiments, the inner layer and the outer layer are formed from a single sheet,

with the inner layer being created by applying a coating to a surface of the sheet, and the outer layer being created from the uncoated portion of the sheet.
(15) Some embodiments further comprise an inner comfort layer that is configured to be positioned to be closer to the skin than the inner layer, wherein the inner comfort layer has a higher wettability than the inner layer. In some embodiments, the inner comfort layer absorbs high surface tension liquids.
(16) Some embodiments further comprise an outer rinsable layer that is configured to be positioned to be further from the skin than the outer layer, wherein the outer rinsable layer has a lower wettability than the outer layer. In some embodiments, the outer rinsable layer repels high surface tension liquids.
(17) Some embodiments further comprise a carbon layer that is configured to absorb vapour, wherein the carbon layer is positioned to be closer to the skin than the outer layer. In some embodiments, the carbon layer comprises activated carbon.
(18) According to some embodiments, a difference in the capillary contact angle between the outer absorption layer and the inner resistant layer is at least 50°. In some embodiments, the difference in the capillary contact angle between the outer absorption layer and the inner resistant layer is at least 60°. In some embodiments, the difference in the capillary contact angle between the outer absorption layer and the inner resistant layer is at least 65°.
(19) According to some embodiments, the garment is configured to cover at least one of a torso, an arm, a leg, a face, a hand or a foot of the wearer when the garment is being worn.
(20) In some embodiments, the garment comprises at least one of a coverall, a jacket, a pair of pants, a vest, a facemask, a glove or a sock.
(21) Some embodiments relate to a layered fabric for resisting penetration of a liquid, the fabric comprising: an outer layer; and an inner layer; wherein the inner layer is configured to be positioned closer to the skin of a wearer than the outer layer when the fabric is being worn; and wherein each of the inner layer and the outer layer on their own are wettable by the liquid; wherein the inner layer has a lower wettability than the outer layer; and wherein the configuration of inner and outer layers creates a wettability difference that biases the outer layer to absorb the liquid and the inner layer to resist penetration of the liquid in a direction moving from the outer layer to the inner layer.
(22) According to some embodiments, when worn with the inner layer positioned closer to the skin of the wearer than the outer layer, the fabric provides greater resistance to penetration of the liquid toward the skin of the wearer than either a fabric made from either the inner layer or the outer layer on its own, or than a fabric worn with the outer layer positioned closer to the skin of a wearer than the inner layer.
(23) According to some embodiments, the liquid is a low surface tension liquid. In some embodiments, the surface tension of the liquid is less than 50 mMm.sup.−1. In some embodiments, the surface tension of the liquid is less than 40 mMm.sup.−1. According to some embodiments, the surface tension of the liquid is less than 30 mMm.sup.−1.
(24) In some embodiments, the garment is air-permeable.
(25) According to some embodiments, the outer layer and the inner layer both repel high surface tension liquids. According to some embodiments, the outer layer and the inner layer both absorb high surface tension liquids. In some embodiments, the inner layer repels high surface tension liquids. In some embodiments, the outer layer absorbs high surface tension liquids. According to some embodiments, the absorption layer and the resistant layer both comprise at least one of a hydrophobic or super hydrophobic material.
(26) According to some embodiments, the inner layer comprises a coating. In some embodiments, the outer layer comprises a coating. In some embodiments, the coating is applied using a dip coating method. According to some embodiments, the coating is applied using a spray coating method. In some embodiments, the inner layer and the outer layer are formed from a single sheet,

with the inner layer being created by applying a coating to a surface of the sheet, and the outer layer being created from the uncoated portion of the sheet.
(27) Some embodiments further comprise an inner comfort layer that is configured to be positioned to be closer to the skin than the inner layer, wherein the inner comfort layer has a higher wettability than the inner layer. In some embodiments, the inner comfort layer absorbs high surface tension liquids.
(28) Some embodiments further comprise an outer rinsable layer that is configured to be positioned to be further from the skin than the outer layer, wherein the outer rinsable layer has a lower wettability than the outer layer. In some embodiments, the outer rinsable layer repels high surface tension liquids.
(29) Some embodiments further comprise a carbon layer that is configured to absorb vapour, wherein the carbon layer is positioned to be closer to the skin than the outer layer. In some embodiments, the carbon layer comprises activated carbon.
(30) According to some embodiments, a difference in the capillary contact angle between the outer absorption layer and the inner resistant layer is at least 50°. In some embodiments, the difference in the capillary contact angle between the outer absorption layer and the inner resistant layer is at least 60°. In some embodiments, the difference in the capillary contact angle between the outer absorption layer and the inner resistant layer is at least 65°.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) Embodiments are described in further detail below, by way of example and with reference to the accompanying drawings, in which:
(2) FIG. **1** shows a previously known layered fabric;
(3) FIG. **2** shows liquid passing through the fabric of FIG. **1**;
(4) FIG. **3** shows a layered fabric according to some embodiments;
(5) FIG. **4** shows a liquid passing through the fabric of FIG. **3**;
(6) FIG. **5** shows a graph comparing the liquid capacity of a range of layered fabrics having different contact angles; and
(7) FIG. **6** shows a garment made from the fabric of FIG. **3**;
(8) FIG. **7** shows a layered fabric according to some alternative embodiments; and
(9) FIG. **8** shows a layered fabric formed from a single sheet according to some further embodiments.
DETAILED DESCRIPTION
(10) Embodiments generally relate to fabrics and garments that are resistant to liquid penetration, as well as to methods for the production of such fabrics and garments. In particular, described embodiments relate to fabrics and materials that use the principle of wettability competition to improve the effectiveness of the protection provided by the fabrics and materials against liquid penetration.
(11) Air and moisture permeable fabrics are widely used for chemical protection. Air and moisture permeable fabrics allow a convective flow of air and moisture around the body, operating as a breathable barrier. However, existing fabrics used for permeable chemical protective clothing (CPC) do not perform well at repelling liquids with a low surface tension. Low surface tension liquids may include ethanol, 2-chloroethyl phenyl sulfide, diethyl chlorophosphate, triethyl phosphate, and diisopropyl methylphosphonate. According to some embodiments, a low surface tension liquid may be any liquid with a surface tension of less than 50 mNm.sup.−1. According to some embodiments, a low surface tension liquid may be any liquid with a surface tension of less than 40 mNm.sup.−1. According to some embodiments, a low surface tension liquid may be any

liquid with a surface tension of less than 30 mNm.sup.−1.
(12) Described embodiments are designed to exploit a synergistic effect or wettability competition mechanism. Specifically, the wettability competition mechanism relates to the phenomenon whereby when placing a lower wettability layer underneath a higher wettability layer within a fabric, the fabric will resist liquid penetration in the direction moving from the high wettability layer to the low wettability layer, even where each layer on its own is wettable by or susceptible to penetration by target liquids. This phenomenon is particularly seen in cases where the difference in wettability between the two layers is high.
(13) While the terms high wettability and low wettability are used, it is to be understood that these terms are relative to each other, and that it is the difference in wettability between the layers that allows for fabrics to resist penetration by a target liquid. Both layers may be wettable by the target liquid, which may be a low surface tension liquid.
(14) FIG. **1** shows a previously used layered fabric **100** for resisting liquid penetration. Layered fabric **100** comprises an outer layer **110** comprising a sheet of material having a low wettability. Outer layer **110** may be an outer layer in that it may be designed to be placed away from the skin of a user or the product that is being protected by fabric **100**, and is instead exposed to the environment in use. Wettability is a measure of how well a liquid wets or adheres to a solid, so a material having a low wettability will not absorb liquid easily. Liquid dropped onto a low wettability material will tend to be suspended above the material with a high contact angle for a longer period of time. For example, a small amount of low surface tension liquid such as 10 μL ethanol dropped onto a low wettability material such as coated polyester may have an initial contact angle of 115° and may hold this contact angle for around 30 seconds before being absorbed.
(15) Layered fabric **100** further comprises an inner layer **120** comprising a sheet of material having a high wettability. Inner layer **120** may be an inner layer in that it is designed to be placed against or adjacent to the skin of a user or a product that is being protected by fabric **100** in use. A material having a high wettability will tend to absorb liquid easily. Liquid dropped onto a high wettability material will tend to be absorbed into or spread over the material, exhibiting a higher contact angle for a shorter period of time. For example, a small amount of low surface tension liquid such as 10 μL ethanol dropped onto a high wettability material such as uncoated polyester may have an initial contact angle of 20° and may hold this contact angle for less than 1 second before being absorbed.
(16) While layers **110** and **120** may both be wettable by low surface tension liquids and may both have a static contact angle (CA) of 0° for low surface tension liquids in some embodiments, the difference in the amount of time it takes for a liquid to spread over the surface of layers **110** and **120** shows that they have differing wettabilities to the liquid.
(17) When a liquid comes in contact with layer **110**, the low wettability of layer **110** resists absorption of the liquid, resulting in the liquid tending to remain on the surface of layer **110** rather than being absorbed. However, a liquid with a low surface tension will eventually be absorbed by layer **110**, especially after being exposed to layer **110** for a long duration. As the liquid passes through layer **110**, it is then quickly absorbed by layer **120**. However, as described below, layered fabric **100** may still allow penetration of liquid in some circumstances.
(18) FIG. **2** shows the path of the liquid **220** with respect to a layered fabric **100** that is in contact with a wearer's skin **210**.
(19) Liquid **220** starts by dropping onto the surface of layer **110**. As described above, the liquid **220** may eventually penetrate through layer **110**, as shown by penetrated liquid **230**, especially where liquid **220** is a liquid with a low surface tension. Some of penetrated liquid **230** is absorbed along the direction shown by arrows **240**, along the length of layer **120** of fabric **100**. However, some of penetrated liquid **230** will eventually penetrate through layer **120** along the direction of arrow **250**, especially after a long duration of time has passed or a high volume of liquid **220** has been penetrated. This becomes liquid **260**, which is free to contact skin **210**. Fabrics such as fabric **100** promote liquid penetration toward skin **210**, as the layer in contact with the skin, being layer **120**,

has higher wettability.
(20) An example of a low surface tension liquid is pure ethanol. An experiment done by dropping 200 μL ethanol onto a dual layer fabric similar to fabric **100** showed that the ethanol slowly spread on the top of the fabric, then moved freely through the fabric, and quickly spread to the bottom layer, as illustrated in FIG. **2**. From the outer surface to the inner surface, the ethanol penetrated through the fabric in 0.72 seconds.
(21) Attempts have been made to improve the liquid protection capability of fabrics such as layered fabric **100** by enhancing the liquid repellency, hydrophobicity or oleophobicity of the lower wettability layer **110**, and/or by increasing absorption or hydrophilicity of the higher wettability layer **120**. For example, coating techniques using fluorine-containing chemicals and nanomaterials have been used. However, it is difficult to prevent penetration of low surface tension liquids, such as low surface chemical warfare agents, using a superoleophobic material. Prior methods for increasing the liquid repellency may also decrease the breathability of fabric **100**, and may increase the thickness and weight of fabric **100** in some cases.
(22) FIGS. **3**, **4**, **7** and **8** illustrate alternative layered fabrics **300**, **700** and **800** that operate in a different way. Instead of concentrating on improving the inherent functions of each layer, fabrics **300**, **700** and **800** use the interaction between fabric layers to improve the prevention of liquid penetration. As described in further detail below, fabrics **300**, **700** and **800** may provide an improved resistance to liquid penetration compared to fabric **100**, especially for liquids having a low surface tension. Fabrics **300**, **700** and **800** may be designed to exploit a synergistic effect or wettability competition mechanism. Specifically, the wettability competition mechanism relates to the phenomenon whereby when placing a lower wettability layer underneath a higher wettability layer within the fabric, the fabric will resist liquid penetration in the direction moving from the high wettability layer to the low wettability layer, even where each layer on its own is wettable by or susceptible to penetration by low surface tension liquids. This phenomenon is particularly seen in cases where the difference in wettability between the two layers is high.
(23) FIG. **3** shows layered fabric **300** in further detail. Layered fabric **300** comprises an absorbent layer **310** comprising a sheet of material having a high wettability, and a resistant layer **320** comprising a sheet of material having a low wettability. According to some embodiments, fabric **300** is a porous and/or permeable fabric.
(24) While the terms high wettability and low wettability are used, it is to be understood that these terms are relative to each other, and that it is the difference in wettability between the layers that allows for fabric **300** to resist penetration by a target liquid. Each of layers **310** and **320** may be wettable by the target liquid on their own, which may be a low surface tension liquid. However, the configuration of the layers in the manner described creates a wettability difference that biases the outer layer **310** to absorb the liquid and the inner layer **320** to resist penetration of the liquid in a direction moving from the outer layer **310** to the inner layer **320**, as described in further detail below.
(25) The wettability of layer **310** is higher than the wettability of layer **320**. Absorbent layer **310** is positioned on the outer side of resistant layer **320**, so that resistant layer **320** is positioned to be closer to the skin **210** of a wearer when the wearer is wearing fabric **300**. In some embodiments, layer **310** may be a PET or other polyester fabric, a wool, a cotton, a nano-spun fabric such as a nanofiber, or a blend such as a Modacrylic-cotton, for example. When worn with layer **320** positioned closer to the skin of the wearer than the layer **310**, fabric **300** or a garment made of such fabric provides greater resistance to penetration of the liquid toward the skin of the wearer than either a fabric made from either the layer **310** or layer **320** on its own, or than a fabric or garment worn with the layer **310** positioned closer to the skin of a wearer than layer **320**.
(26) According to some embodiments, both layers **310** and **320** may be hydrophobic and/or superoleophobic. According to some embodiments, both layers **310** and **320** may have a low wettability to high surface tension liquids, such as water or oil. However, as noted above, both

layers **310** and **320** may be wettable by low surface tension liquids, such as ethanol, for example. Resistant layer **320** may comprise a hydrophobic or super hydrophobic material in some embodiments. Resistant layer **320** may have a low wettability to high surface tension liquids. In some alternative embodiments, both layers **310** and **320** may be hydrophilic, or have a high wettability to high surface tension liquids. Absorbent layer **310** may comprise a hydrophilic or super hydrophilic material in some embodiments, or have a high wettability to high surface tension liquids. It is noted that hydrophobia and hydrophilia specifically relate to the relationship between a material and water, while wettability may refer to any liquid.

(27) In some embodiments, layer **320** may comprise a material having a coating, such as a polyester fabric having a hydrophobic coating. For example, layer **320** may comprise a polyester or cotton fabric coated with one or more of FD-POSS, tridecafluorooctyl triethoxysilane (FAS) or PVDF-HFP using a dip coating method. According to some embodiments, the coating may comprise 100 ml acetone containing 1.0 g of FD-POSS and 5.0 g of FAS, or similar ratios, for example. According to some embodiments, the coating may comprise 100 ml acetone containing 0.7 g of FD-POSS and 3.0 g of FAS, or similar ratios, for example. According to some embodiments, the coating may comprise 100 ml of a solvent mixture of 50:50 vol/vol DMF and acetone containing 16 g of PVDF, or similar ratios, for example. According to some embodiments, layer **320** may comprise Nomex. Where layer **320** comprises a coated fabric, the coating may be between 30 nm and 50 nm thick in some embodiments.

(28) According to some embodiments, the coating may be applied to layer **320** by directly applying a coating solution to layer **320** using a dip coating method. According to some embodiments, layer **320** may be immersed in the coating solution for at least 5 minutes, at least 10 minutes, or at least 20 minutes. According to some embodiments, the dip coated layer **320** may be squeezed to remove the extra solution, which may be done with a rubber roller, for example. Layer **320** may be dried at room temperature. According to some embodiments, layer **320** may alternatively or additionally be dried at a higher temperature, such as at 135°, for example. The higher temperature drying may be performed for a duration of around 30 minutes in some embodiments. According to some embodiments, the coating solution may be 100 ml acetone containing 1.0 g of FD-POSS and 5.0 g of FAS. In some embodiments, the coating solution may be acetone containing a different concentration of FD-POSS and FAS.

(29) According to some embodiments, layer **310** may comprise a coating, which may be a coating such as that described above with respect to layer **320**.

(30) Layers **310** and **320** may each have a weight of around 150 to 170 g/m.sup.2. Each of layers **310** and **320** may be around 450 to 6001 μm in thickness.

(31) The wettability of a layer of fabric **300** may be measured based on the capillary wall contact angle of the layer, according to the Lucas-Washburn equation, as described below with reference to FIG. **5**.

(32) While fabric **300** comprises two sheets of material **310** and **320**, some embodiments relate to a fabric **800**, as shown in FIG. **8**. Fabric **800** comprises a single sheet of material **805** which starts as a single layer and is subject to processes that change the properties of sections of fabric **800**, causing fabric **800** to develop layers having different behaviors. For example, layer **820** may be developed by applying a coating to one surface of fabric **800**, while layer **810** may be the remaining uncoated part of fabric **800**. The configuration of the layers **810** and **820** in the manner described creates a wettability difference that biases the outer layer **810** to absorb the liquid and the inner layer **820** to resist penetration of the liquid in a direction moving from the outer layer **810** to the inner layer **820**, as described above with reference to layers **310** and **320** of FIG. **3**.

(33) According to some embodiments, the coating applied to fabric **800** may be applied by spray coating. For example, according to some embodiments the coating may be formed by spraying FD-POSS/FAS on one side of fabric **800** using a purpose-built spraying device. The coating thickness may be controlled by adjusting the amount of the coating solution to cause around the half of

thickness of fabric **800** to be coated. After applying the coating, fabric **800** may be dried at room temperature. According to some embodiments, fabric **800** may alternatively or additionally be dried at a higher temperature, such as at 135°, for example. The higher temperature drying may be performed for a duration of around 30 minutes in some embodiments.

(34) In some alternative embodiments, layers **810** and **820** may each be developed by applying different coatings to the opposing surfaces of fabric **300**.

(35) FIG. **7** shows an alternative layered fabric **700**. Layered fabric **700** comprises a high wettability or absorbent layer **310** and a low wettability or resistant layer **320**, which have the same properties as described above with reference to FIG. **3**. According to some embodiments, fabric **700** may alternatively comprise layers **810** and **820** which are formed from a single sheet to which at least one coating is applied, as described above with reference to FIG. **800**.

(36) Fabric **700** may also comprise further, optional layers. In the illustrated embodiment, Fabric **700** also comprises an optional carbon layer **340** While the illustrated embodiment shows carbon layer situated between layer **310** and layer **320**, in some embodiments carbon layer **240** may be situated on the inside of layer **320**. Carbon layer **340** may be made of activated carbon, and may be designed to absorb chemical vapour, resisting the penetration of chemical vapour through penetrating fabric **300**.

(37) Fabric **700** may also comprise an optional outer layer **330**, which may comprise a low wettability layer. Outer layer **330** may be hydrophobic or super hydrophobic in some embodiments. Outer layer **330** may have a lower wettability than absorbent layer **310**. According to some embodiments, outer layer **330** may comprise the same or similar material to layer **320**. The inclusion of layer **330** may allow for fabric **300** to repel liquids with high surface tension, which may allow fabric **700** to be able to be rinsed, which may be particularly important if the outer surface of fabric **700** has been exposed to harmful liquid chemicals. Outer layer **330** may therefore be a rinsable or washable layer in some embodiments.

(38) Fabric **700** may also comprise an optional inner layer **350**, which may comprise a high wettability layer. Inner layer **350** may be hydrophilic or super hydrophilic in some embodiments. Inner layer **350** may have a higher wettability than resistant layer **320**. According to some embodiments, inner layer **350** may comprise the same or similar material to layer **310**. According to some embodiments, layer **310** may have a higher wettability than layer **350**. The inclusion of layer **350** may provide added comfort and wearability to fabric **350**, by providing a soft and absorbent layer next to the skin **210** of the user, to absorb any moisture such as sweat from skin **210**. Inner layer **350** may therefore be a comfort layer in some embodiments.

(39) While fabric **700** is described as a layered fabric, in some embodiments fabric **700** may start as a single layer and be subject to processes that change the properties of sections of fabric **700**, causing fabric **700** to develop layers having different behaviors. For example, layer **320** may be developed by applying a coating to fabric **700**, while layer **310** may be the remaining uncoated part of fabric **700**. In some embodiments, layers **310** and **320** may each be developed by applying different coatings to the opposing surfaces of fabric **700**.

(40) FIG. **4** shows the path of the liquid **420** with respect to a layered fabric **300** or **700** that is in contact with a wearer's skin **210**. The layered fabric **300** or **700** as illustrated in FIG. **4** comprises only layers **310** and **320**. According to some embodiments, one or more of layers **330**, **340** and **350** may also be included.

(41) Liquid **420** starts by dropping onto the surface of layer **310**. Layer **310** absorbs the liquid by in-plane diffusion, forming absorbed liquid **430** and preventing layer **320** from contacting the liquid. Liquid **430** is absorbed along the direction shown by arrows **440**, along the length of layer **110** of fabric **300** or **700**. Some of absorbed liquid **430** will also be absorbed along the direction shown by arrow **450**. However, this liquid contacts layer **320**, which resists absorption of the liquid **430**. As layer **320** will take a long time to absorb liquid **420**, the liquid will instead continue to be absorbed by layer **310**, resulting in the fabric **300**/**700** tending to resist penetration of liquid **420**

until layer **310** has reached its absorption capacity.
(42) While the above is described with reference to layers **310** and **320** of fabric **300** or **700**, it should be appreciated that the same effect is achieved with layers **810** and **820** of fabric **800**.
(43) When an experiment was done by dropping 200 μL ethanol onto a dual layer fabric similar to fabric **300**, the ethanol rapidly spread on the outer layer **310** without penetrating to reach the inner layer **320**. As a result, inner surface **320** remained dry. This result suggests that fabric **300** shows different wicking performance between the two surfaces of layers **310** and **320** compared to fabric **100**, and this wicking prevents liquid penetration when liquid is fed from layer **310** having a higher wettability.
(44) Specifically, in the experiment a polyester fabric (marked "A") and a polyester fabric coated with a polymer solution (marked "B") were tested for wettability using pure ethanol as a low surface tension liquid. Both the uncoated fabric A and coated fabric B showed similar surface morphology because the coating was very thin, at −50 nm. Fabrics A and B were both were wetted by ethanol with a static contact angle of 0°. When 10 μL of ethanol was dropped on sample A, its contact angle changed from 20° to zero in less than 1 second. For fabric B, the ethanol contact angle reduced from 115° to zero in a much longer period (30 seconds). The shorter spread time for A than B suggests that the uncoated PET had higher wettability to ethanol than the coated fabric.
(45) The fabrics A and B were laminated together to get four possible combinations of a dual-layered fabric, being A/A, A/B, B/A, and B/B. Following this, ethanol was dropped to the laminated fabric assemblies to observe its spreading on the fabrics. When 200 ethanol was dropped onto the dual-layer fabric at the B surface of fabric B/A, the ethanol spread on the top B layer then moved freely through the fabric and quickly spread to the bottom A layer. It took 0.72 seconds for ethanol to transport from the feeding to the back surface of layer A. When the same volume of ethanol was fed onto the A surface of fabric A/B, the ethanol only spread on the A layer without spreading into the layer B behind. As a result, the B surface remained dry in the entire process. For the dual-layer fabrics made of two layers of the same fabric, being A/A and B/B, similar wicking performance to the single fabric samples were observed, and these dual-layer fabrics were unable to prevent ethanol penetration. The A/A fabric showed a shorter penetration time (0.08 seconds) than the B/B fabric. This can be explained by the lower ethanol wettability for fabric B than fabric A.
(46) These results indicate that the wettability difference between the fabric layers plays a key role in preventing ethanol penetration across the dual-layer fabric. The A/B dual-layer fabrics show different wicking performances depending on the combination configuration, and ethanol penetration can be prevented when the liquid is fed from the layer with higher wettability.
(47) The maximum volume of ethanol that can be blocked by the dual-layer A/B fabric was also measured by dropping ethanol on one spot of the A surface until the ethanol started penetrating across the two-fabric layer. The maximum absorption capacity was found to depend mainly on the absorption capacity of fabric A and the wettability difference between fabrics A and B. The absorption capacity of fabric A may depend on factors such as the fabrics structure and the viscosity of the liquid. If the time taken to reach the absorption capacity of layer A is less than the time required to start wetting layer B, layer A will be fully loaded before the ethanol starts to travel to layer B resulting in the maximum volume being absorbed. Otherwise, if the time taken to reach the absorption capacity of layer A is greater than the time taken to begin wetting layer B, the ethanol will begin to penetrate layer B before reaching the maximum capacity of layer A, hence not reaching the maximum volume.
(48) The maximum volume absorbed by fabric A/B was 220 μl. The ethanol dropped covered 8.12 cm.sup.2 area of the A surface. The diameter of ethanol spread on the top surface was approximately 3.22 cm. Therefore, the volume of the ethanol per unit area of the fabric A/B to prevent liquid penetration was 27.09111/cm.sup.2. Based on this data, one can estimate the volume of ethanol that can be blocked using a specific configuration of the fabric, assuming that the

ethanol is uniformly applied to the A-side of the fabric. When dropping ethanol on the same thickness of the B/A fabric but applied to the B side, the maximum volume absorbed was only 3 μl, which is 70 times less than for fabric A/B. For the A/A and B/B dual layer fabrics, the maximum volumes absorbed were less than 2 μl. Apart from ethanol, other low surface tension liquids, such as 2-chloroethyl phenyl sulfide, diethyl chlorophosphate, triethyl phosphate, and diisopropyl methylphosphonate were tested using the dual fabrics A/B, A/A, and B/B, and similar wicking performances were observed.
(49) By increasing the wettability difference between layers **310** and **320** or **810** and **820**, it may be possible to increase the capacity of fabric **300**, **700** or **800** in preventing liquid penetration by 200 to 1000 times compared to fabric **100**, while maintaining the same or even a lower thickness and weight, and keeping the same or even a higher flexibility, heat transfer and air-permeability. As well as being used for CPC and other liquid protection garments, fabric **300**, **700** and **800** may be used for preventing liquid penetration in other applications, such as applications relating to hygiene, food preparation, soil water retention and sports clothing, for example.
(50) FIG. **5** shows a graph **500** showing the effect of capillary contact angle difference on the liquid capacity of a number of dual layer fabrics. In contrast to the contact angle, which describes the angle that a liquid creates with a flat solid surface, a capillary contact angle refers to the contact angle between a liquid and a tube wall, or capillary. The capillary contact angle relates to the ability of the liquid to fill small capillaries or pores, such those inside or between the fibres of a fabric.
(51) Graph **500** has an x-axis **510** showing a contact angle difference between the layers of a number of tested fabrics, Δθ=θ.sub.β−θ.sub.τ, Graph **500** further has a y-axis **520** showing a maximum capacity in μL for each of the tested fabrics. Each dual layer fabric was tested for the volume of ethanol it could hold, and the results were recorded and graphed.
(52) 18 dual layer fabrics were tested. The two layers of each dual layer fabric were selected from polyester (marked "A"), and five types of coated polyester (marked "B1", "B2", "B3", "B4" and "B") having five different concentrations of FD-POSS/FAS to produce five different wettabilities. Each of the tested layers were wettable to ethanol and had a static contact angle of 0° for ethanol.
(53) The corresponding wicking height for each layer was measured, and the results are reproduced below in Table 1. The time taken for each layer to wick 1 cm was also recorded. For layer B, it took 195 seconds for the ethanol to wick 1 cm up, which was 192.4 seconds slower than it took the ethanol to reach the same height on layer A. Layer B also showed a significantly lower total wicked height compared to layer A. While layer A was able to wick ethanol to a height of 9 cm, layer B was 7 cm lower, with a maximum height of 2 cm.
(54) TABLE-US-00001 TABLE 1 maximum wicking height of each tested layer A B1 B2 B3 B4 B Max Wicking Height (mm) 90 88 61 44 45 20
(55) The wicking times for each fabric to reach a wicking height of 10 mm were also recorded, and are reproduced below in Table 2.
(56) TABLE-US-00002 TABLE 2 maximum wicking height of each tested layer A B1 B2 B3 B4 B Wicking time (s) 2.56 2.6 5.6 13 14 195
(57) Layers A, B, B1, B2, B3 and B4 were fabricated to create a series of dual layer fabrics, and the maximum liquid capacity of the dual layer fabrics was also measured.
(58) The wicking height L achieved in a time t can be described for a particular fabric with the Lucas-Washburn equation,
(59) $L = \sqrt{\frac{rt\cos(\ )}{2}}$, where φ is the contact angle between the liquid and the idealized capillary wall within the fabric, r is the radius of the idealized capillary, η is the dynamic viscosity of the liquid, and γ is the surface tension.
(60) Using the Lucas-Washburn equation, the contact angle difference between two fabrics or two fabric layers can be calculated based on the wicking height of each fabric or layer.
(61) Since η, γ, and r are almost identical between the samples A, B1, B2, B3, B4 and B, the

difference in wicking height is only determined by the contact angle. Therefore, the wicking height and contact angle for different samples have the following relationship:
(62) $\cos{}_1 = \frac{L_1^2}{L_2^2} \times \cos{}_2$ where L.sub.1 and θ.sub.1 are the maximum wicking height and capillary contact angle of the first sample, and L.sub.2 and θ.sub.2 are the maximum wicking height and capillary contact angle of the second sample.
(63) This can also be determined using the relationship between the wicking time and contact angle for different samples, being:
(64) $\cos{}_1 = \frac{t_2}{t_1} \times \cos{}_2$ where t.sub.1 and θ.sub.1 are the wicking time to reach wicking height of 10 mm and capillary contact angle of sample 1; t.sub.2 and θ.sub.2 are the wicking time to reach wicking height of 10 mm and capillary contact angle of sample 2.
(65) A smooth PET film was used as the uncoated layer A, with a capillary contact angle of 11.08°. Based on the above equations and according to the wicking height and time data, the capillary contact angles for the other samples were obtained based on the above equations, as listed below in Table 3.
(66) TABLE-US-00003 TABLE 3 calculated capillary contact angles for each dual layer fabric Bottom Average Sample Top θ θ Δθ (ul) A/A 11.08 11.09 0 1.00 A/B1 11.08 20.24 9.16 16.67 A/B2 11.08 63.20 52.12 33.33 A/B3 11.08 75.79 64.72 41.67 A/B4 11.08 76.43 65.36 38.33 A/B 11.08 87.22 76.14 220.00 B1/A 20.24 11.08 −9.16 1.00 B1/B1 20.24 20.24 0 6.67 B1/B2 20.24 63.20 42.96 16.67 B1/B3 20.24 75.79 55.56 31.67 B1/B4 20.24 76.43 56.19 3500 B1/B 20.24 87.22 66.97 106.67 B2/A 63.20 11.08 −52.12 1.00 B2/B1 63.20 20.24 −42.96 1.00 B2/B2 63.20 63.20 0 11.67 B2/B3 63.20 75.79 12.59 8.33 B2/B4 63.20 76.43 13.23 11.67 B2/B 63.20 87.22 24.019 36.67
(67) For fabrics A and B, the difference in contact angle AO was calculated to be 76.1°.
(68) FIG. **5** shows the relationship between the maximum liquid capacity and the difference in contact angle for the tested fabrics. Plot **530** shows dual layer fabrics having layer A as the top fabric, plot **540** shows dual layer fabrics having layer B1 as the top fabric, and plot **550** shows dual layer fabrics having layer B2 as the top fabric. Larger differences in contact angle led to higher maximum capacities, resulting in the improved fabric as described above with reference to FIGS. **3**, **4**, and **7**. This phenomenon is described in further detail below with reference to capillary force within the yarn fibres of a fabric.
(69) As seen from FIG. **5**, an increase in capacity is seen in dual layer fabrics having a capillary contact angle difference of at least 50°, and a significant increase in capacity is seen in dual layer fabrics having a capillary contact angle difference of at least 65°. Based on these results, fabric **300**, **700** or **800** may be manufactured so that the contact angle difference between layers **310** and **320** or **810** and **820** is at least 50°. According to some embodiments, fabric **300**, **700** or **800** may be manufactured so that the contact angle difference between layers **310** and **320** or **820** and **820** is at least 60°. In some embodiments, fabric **300**, **700** or **800** may be manufactured so that the contact angle difference between layers **310** and **320** or **810** and **820** is at least 65°. According to some embodiments, fabric **300**, **700** or **800** may be manufactured so that the contact angle difference between layers **310** and **320** or **810** and **820** is at least 70°.
(70) A woven fabric made of multifilament yarns has two types of pores within it, being smaller pores within the yarn between fibres, and coarser pores between the adjacent yarns. The pores within the yarn may be referred to as intra-yarn pores, and can be idealized as cylindrical shapes. The pore diameter of the intra-yarn pores is decided by the inter-fibre distance (d). The pores between yarns may be knows as inter-yarn pores, and are generally irregular in size and shape, with their size (D) decided by the yarn diameter and weaving structure.
(71) The capillary force of an idealized cylindrical capillary can be determined by the equation:
(72) $F = \frac{2\gamma\cos\theta}{r}$.
(73) Within a single fabric layer, F is mainly determined by the radius of the pores. Since the

diameter of the intra-yarn pores d is much smaller than the diameter of the inter-yarn pores D, the capillary force within each yarn fibre is significantly larger than that between the yarn fibres. When liquid attaches to fabric, the movement of the liquid along an in-plane direction within each yarn fibre is a lot faster than that along the cross-plane. When two layers of fabrics are laminated, larger pores are created between the layers, slowing down cross-plane wicking. When the second layer has a lower wettability than the first layer, the corresponding capillary forces are smaller. If the liquid dropped in the first layer is entirely consumed by the in-plane wicking, there will be no liquid drawn into the second layer. As a result, no liquid penetration across the second layer happens.
(74) FIG. **6** shows an example protective garment **600** comprising fabric **300**, **700** or **800**. Protective garment **600** may be designed to isolate a wearer from hazardous liquid industrial chemicals (TICs), chemical warfare agents (CWAs) and other harmful liquids, vapours, and aerosols.
(75) Garment **600** comprises an external surface **610** and an internal surface **620**. Internal surface **620** is configured to be positioned closer to the skin of a wearer than external surface **610** when garment **600** is worn. External surface **610** is configured to be exposed to the environment of the wearer. Features of the garment such as pockets, seams and closures may be located taking into account which surface is configured to be the internal surface **620** and which surface is configured to be external surface **610**. For example, closures such as zippers and buttons may be configured to be accessed from external surface **610**. Seams may be sewn such that seam allowances are configured to be on internal surface **620**. Pockets may be located on both internal surface **620** and external surface **610** depending on their location. For example, pockets designed to be located on the lower half of the body, such as on the legs or hips, may be designed to be accessed from external surface **610**. Some pockets located on the upper body may be configured to be accessible from internal surface **620**, such as internal breast pockets.
(76) As illustrated, garment **600** is designed to cover a relatively large proportion of the wearer's body, with garment **600** being a coverall style garment. Garment **600** includes a bodice **630** configured to cover a torso of a wearer. Garment **600** further includes sleeves **640** configured to cover the arms of a user, and pant legs **650** configured to cover the legs of a user.
(77) According to some alternative embodiments, garment **600** may be a different type of garment. For example, garment **600** may be a face mask configured to cover a face of a user; a vest configured to cover the torso of a user; a jacket configured to cover the torso and arms of a user; pants configured to cover the legs of a user; gloves configured to cover the hands of a user; or socks, configured to cover the feet of a user, for example.
(78) According to some embodiments, fabric **300**, **700** or **800** may be cut and sewn to create garment **600**. According to some alternative embodiments, fabric **300**, **700** or **800** may be shaped into garment **600** at the time of manufacture. When used to create garment **600**, fabric **300**, **700** or **800** may be oriented so that layer **350**, **320** or **820** forms internal surface **620**, and so that layer **330**, **310** or **810** forms external surface **610**.
(79) While described embodiments have generally been described as suitable for use as protective clothing, those skilled in the art will recognise that the described embodiments may be used in other products, including baby nappies, hygiene products, wound care, food preparation, soil water retention, and sporting/camping/ski equipment and clothing.
(80) It will be appreciated by persons skilled in the art that numerous variations and/or modifications may be made to the above-described embodiments, without departing from the broad general scope of the present disclosure. The present embodiments are, therefore, to be considered in all respects as illustrative and not restrictive.

Claims

1. A garment for resisting penetration of a target liquid, the garment comprising a dual layer fabric, the dual layer fabric comprising: an outer layer; and an inner layer; wherein both the inner layer and outer layer comprise hydrophobic or superhydrophobic material; wherein the inner layer is to be positioned closer to the skin of a wearer than the outer layer when the garment is being worn; wherein each of the inner layer and the outer layer on their own are wettable by the target liquid; wherein the inner layer has a lower wettability to the target liquid than the outer layer; wherein the surface tension of the target liquid is less than 50 mNm.sup.−1; and wherein the configuration of the inner and outer layers create a wettability difference that biases the outer layer to absorb the target liquid and the inner layer to resist penetration of the target liquid in a direction moving from the outer layer to the inner layer.
2. The garment of claim 1, wherein the surface tension of the target liquid is less than 30 mNm.sup.−1, and both the outer layer and the inner layer are superoleophobic.
3. The garment of claim 1, wherein the surface tension of the target liquid is less than 40 mNm.sup.−1.
4. The garment of claim 1, wherein the surface tension of the target liquid is less than 30 mNm.sup.−1.
5. The garment of claim 1, wherein the garment is air-permeable.
6. The garment of claim 1, wherein the inner layer and the outer layer are formed from a single sheet, with different coatings applied for defining the inner and outer layers respectively; or wherein the inner and outer layers are laminated together.
7. The garment of claim 1, further comprising an inner comfort layer that is to be positioned to be closer to the skin than the inner layer, wherein the inner comfort layer has a higher wettability than the inner layer.
8. The garment of claim 7, wherein the inner comfort layer is hydrophilic or superhydrophilic.
9. The garment of claim 1, further comprising an outer rinsable layer that is to be positioned further from the skin than the outer layer, wherein the outer rinsable layer has a lower wettability than the outer layer.
10. The garment of claim 9, wherein the outer rinsable layer is hydrophobic or superhydrophobic.
11. The garment of claim 1, further comprising a carbon layer that is configured to absorb vapour, wherein the carbon layer is positioned on the inside of the inner layer.
12. The garment of claim 11, wherein the carbon layer includes activated carbon.
13. The garment of claim 1, wherein a difference in the capillary contact angle between the outer layer and the inner layer is at least 50°.
14. The garment of claim 1, wherein a difference in the capillary contact angle between the outer layer and the inner layer is at least 60°.
15. The garment of claim 1, wherein the garment is to cover at least one of a torso, an arm, a leg, a face, a hand and a foot of the wearer when the garment is being worn.
16. The garment of claim 1, wherein the garment includes at least one of a coverall, a jacket, a pair of pants, a vest, a facemask, a glove or a sock.
17. The garment of claim 1, wherein the inner layer comprises a first coating and the outer layer comprises a second coating.
18. The garment of claim 17, wherein the first and second coatings comprise one or more of FD-POSS, tridecafluorooctyl triethoxysilane (FAS), and/or PVDF-HFP.
19. The garment of claim 18, wherein the first and second coatings comprise different concentrations of FD-POSS/FAS.
20. The garment of claim 17, wherein the inner layer and the outer layer comprise coated polyester or cotton fabric.
21. The garment of claim 1, wherein both the inner layer and the outer layer are superhydrophobic.
22. The garment of claim 1, wherein the target liquid is hazardous to the wearer; and/or wherein the

target liquid is a chemical warfare agent.

23. The garment of claim 1, wherein, to create the wettability difference for resisting penetration of the target liquid, the outer layer and inner layer of the dual layer fabric comprise: different concentrations of the same hydrophobic or superhydrophobic material; and/or different hydrophobic or superhydrophobic material.